HELLO WORLD DEV CONFERENCE 2026

別讓 Agent 瞎掰：用 Semantic Layer 打造可信賴的企業 AI

Danny Lin
Solutions Architect, AWS

© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.

Agents with Enterprise data – Top 3 challenges

Top models achieve 86% on academic SQL benchmarks but drop to ~40-59% on realistic enterprise schemas without business context and semantic grounding

1 Schema Ambiguity

Hundreds of tables, overlapping column names, undocumented joins. Agent cannot determine which "customer_id" is canonical.

2 No Shared Metric Definitions

"Revenue," "churn," "margin" calculated differently per team. Agent invents its own logic or picks a stale definition.

3 Missing Business Context

Rules live in people's heads. "Promo ROI excludes returns" cannot be inferred from schema alone.

AI Trust Gap – Inconsistent Answers and Lack of Explainability

Sources: Spider 2.0 benchmark · Promethium "State of Text-to-SQL" (2025) ·

aws

Two Ways to Reach an Answer

Businesses don't run on probability.

HOW AN LLM WORKS

Runs on probability

Predicts the next most-probable token. The output is fluent and often impressive — *"that's powerful"* — but it is a probability distribution, not a guarantee.

When the cost of being wrong is high:

~~Fluent Guessing~~

isn't good enough.

HOW A BUSINESS RUNS

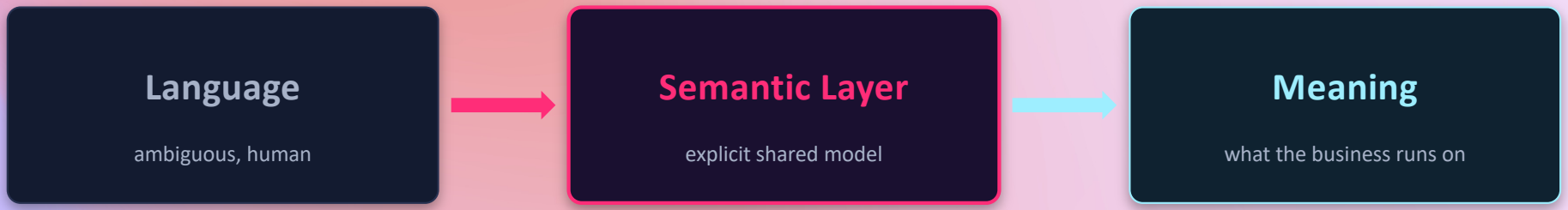
A deterministic chain

1. **Definitions** what each term means
2. **Rules** what must happen
3. **Constraints** what cannot happen
4. **Consequences** what follows

aws

Semantic Layer: The Grounding Layer

You can't automate what you can't clearly articulate.



Semantic layer encodes the same logic the business already runs on — and makes it available to AI:

Definitions | **Rules** | **Constraints** = **grounded AI**

The takeaway: Semantic layer converts the LLM's probable guess into an answer checked against a deterministic model — correct, auditable, and safe to act on when the cost of being wrong is high.

aws

What a Semantic Layer Is?

© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.

Same Data, Two Understandings

What is our promo ROI for Q1 across EMEA, by category?

Agent response:

Q1 EMEA Promo ROI by Category:
Beverages: 3.2x | Snacks: 2.1x | Personal Care: 4.7x

Source: S3 + Nielsen (unified via semantic layer)
Metric: Standardized promo_roi_v3 (governed definition)
Lineage: SAP.promo_spend + Nielsen.incremental_sales

aws

What is Semantic Layer for Agentic AI & Why It Matters

Human practitioners navigate enterprise complexity because they carry institutional understanding — how the business works, where information lives, what rules govern decisions.
AI agents lack this implicit context — they see data but not meaning.

What Practitioners Know

- Business rules and exceptions
- Entity relationships across systems
- Which metrics matter and how they're calculated
- Compliance constraints and boundaries
- Historical context and institutional memory

What Agents See Without Semantic Layer

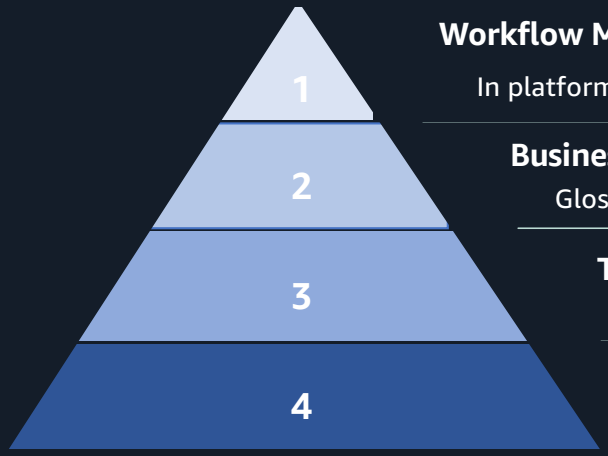
- Raw columns and tables with no meaning
- Disconnected data silos
- No business rules or constraints
- No relationship semantics
- No governed metric definitions

Without shared semantic, every agent independently reconstructs business logic → inconsistency, inaccuracy, and ungovernable outcomes

aws

Consolidated Semantics

Semantics from across the stack available via AWS



Workflow Metadata

In platform prompts/queries/code – Amazon SageMaker Notebooks, Query Editor

Business Metadata

Glossaries, Custom forms, policies – Glue Catalog Business Context

Technical Metadata

Schemas, partitions, lineage - AWS Glue Catalog, Lake Formation

Object Metadata

S3 tags, labels, properties - Amazon S3

SOURCES

Technical & Business Catalog

Glue, Redshift, 3P catalogs

BI Systems

Quick Sight, Tableau, Looker

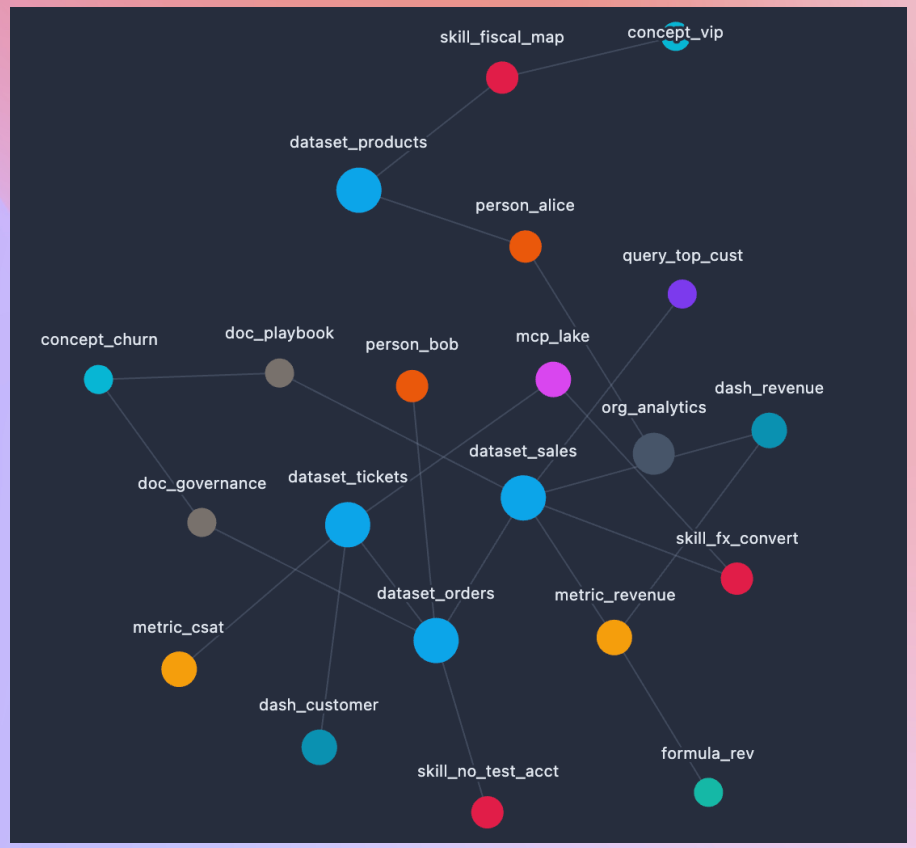
Application, Proprietary Systems

Wikis, logs, ERP systems

© 2026, Amazon Web Services, Inc. or its affiliates.

The knowledge graph: your data estate, connected

Every data asset becomes a navigable node — agents traverse relationships to find what they need



NODE TYPES

- Dataset
- Metric
- Formula
- Skill
- Concept
- Document
- Dashboard
- Query
- Person
- Organization
- MCP Server

What the graph does for agents

Linking heterogeneous data sources together, provide governed data relationships, business rules, and domain knowledge at runtime.

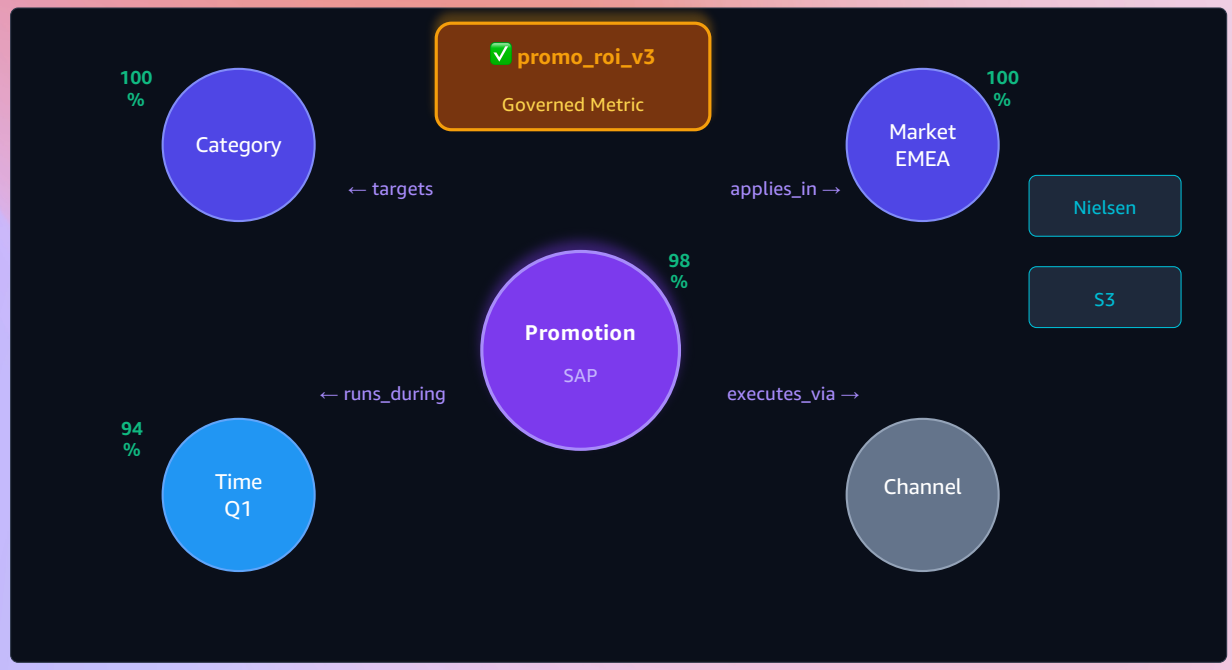
Skills attach to data assets — markdown files with domain-specific instructions (e.g., "Electronics: 30-day returns").

The graph learns from how agents actually use data.

© 2026 Amazon Web Services, Inc. or its affiliates.

Query to Governed Answer — Semantic layer in Action

👤 "What is our promo ROI for Q1 across EMEA, by category?"



Agent Response

Q1 EMEA Promo ROI by Category

Beverages	3.2x
Snacks	2.1x
Personal Care	4.7x

Ontology traversal: 340ms

SOURCES

S3 + Nielsen — unified via semantic layer

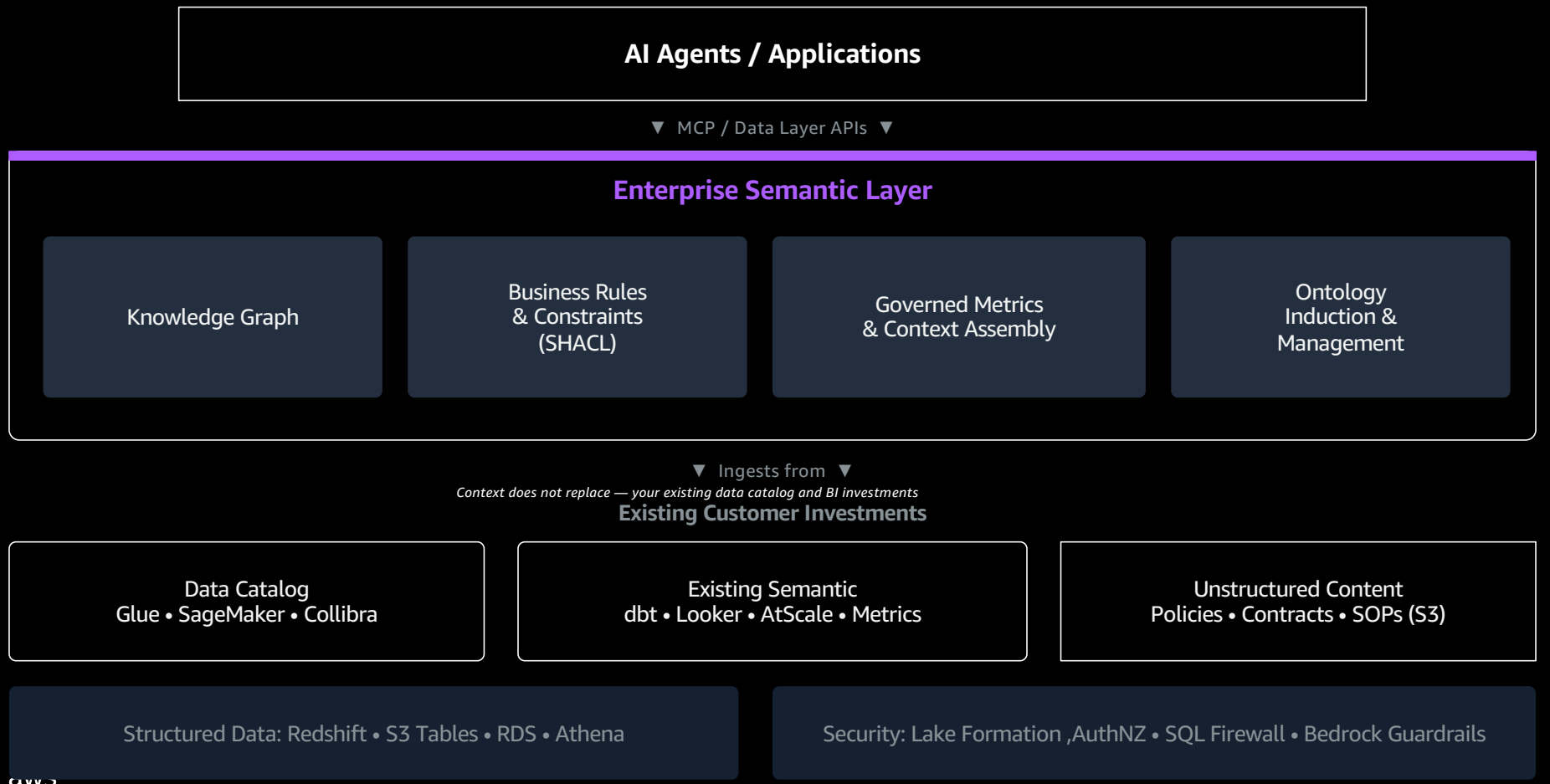
LINEAGE

SAP.promo_spend + Nielsen.incremental_sales → promo_roi_v3

GOVERNANCE

Standardized metric: promo_roi_v3 ✓
Definition enforced across all agents

Technical Architecture: High Level



© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved. Amazon Confidential and Trademark.

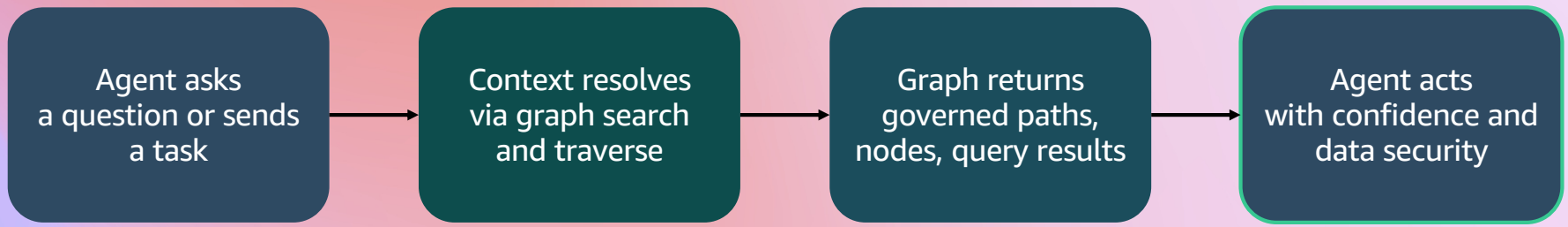


DEMO

© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.

How Agents Use the Graph

One integration serves every agent — no per-app context engineering



Any framework:
- Bedrock AgentCore
- LangChain
- Custom MCP client

- Search knowledge graph
- Resolves entities
- Traverse through edges
- Applies identity scope

Output through API/MCPs
- Low level API for Search and Traverse
- High level API/MCP for agentic query

- Correct answer
- No hallucination
- Auditable trace
- Reusable across agents

API + MCP: agents consume context the same way regardless of platform. Build once, serve every agent.

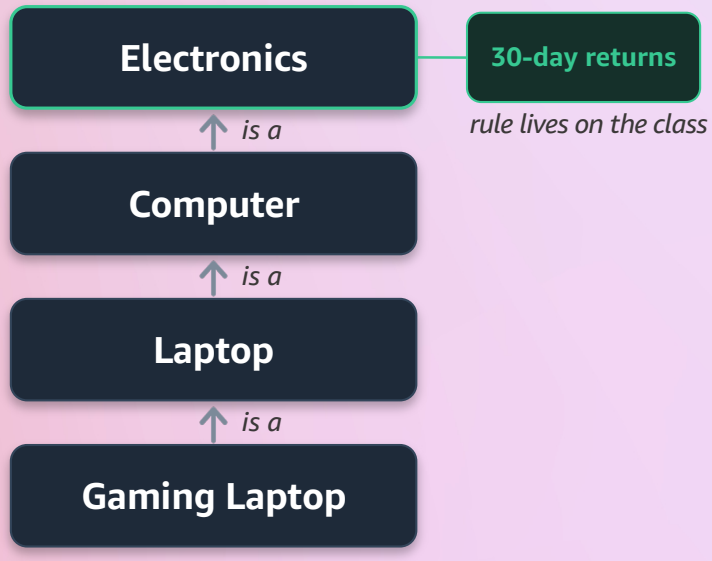
aws

What a formal ontology really is

A formal ontology is a mathematically precise model of a business domain that a machine can reason over. It is defined using formal logic with precise semantics and rigorous rules, commonly represented with standards such as OWL on RDF.

It is not a glossary or a tag list. It states facts such as “every laptop is a computer” and “every computer is electronics,” so a system can follow the chain and draw conclusions.

Classes, relationships, and rules a machine can follow.



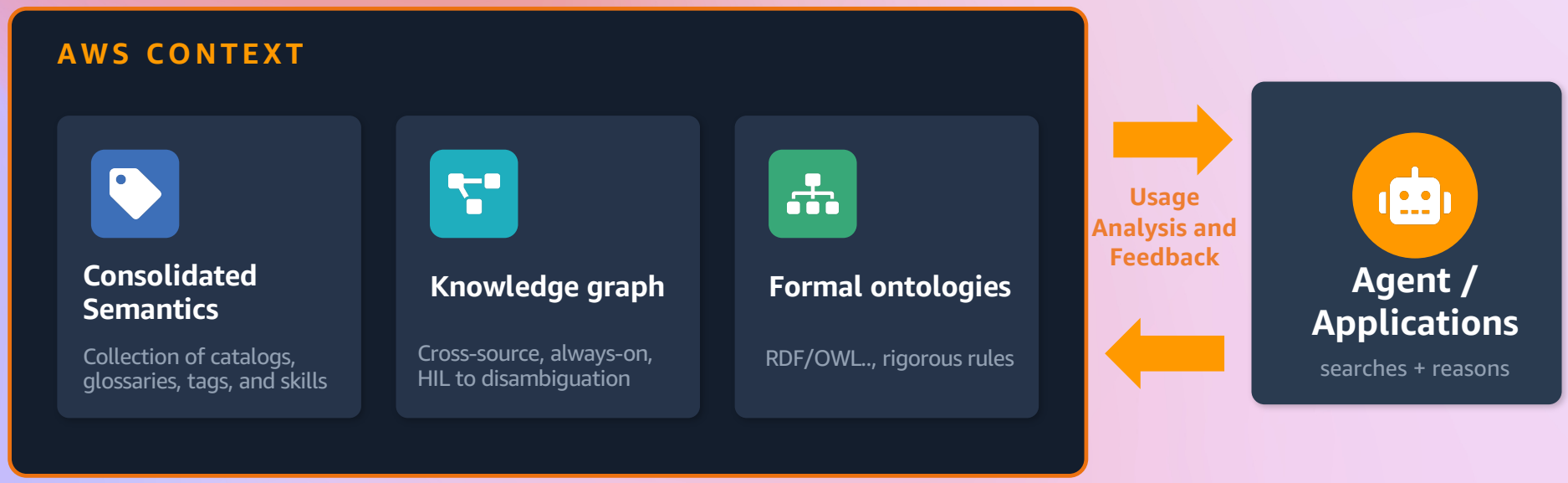
aws

AWS Context: The Full Stack Agents Reason Over

© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.

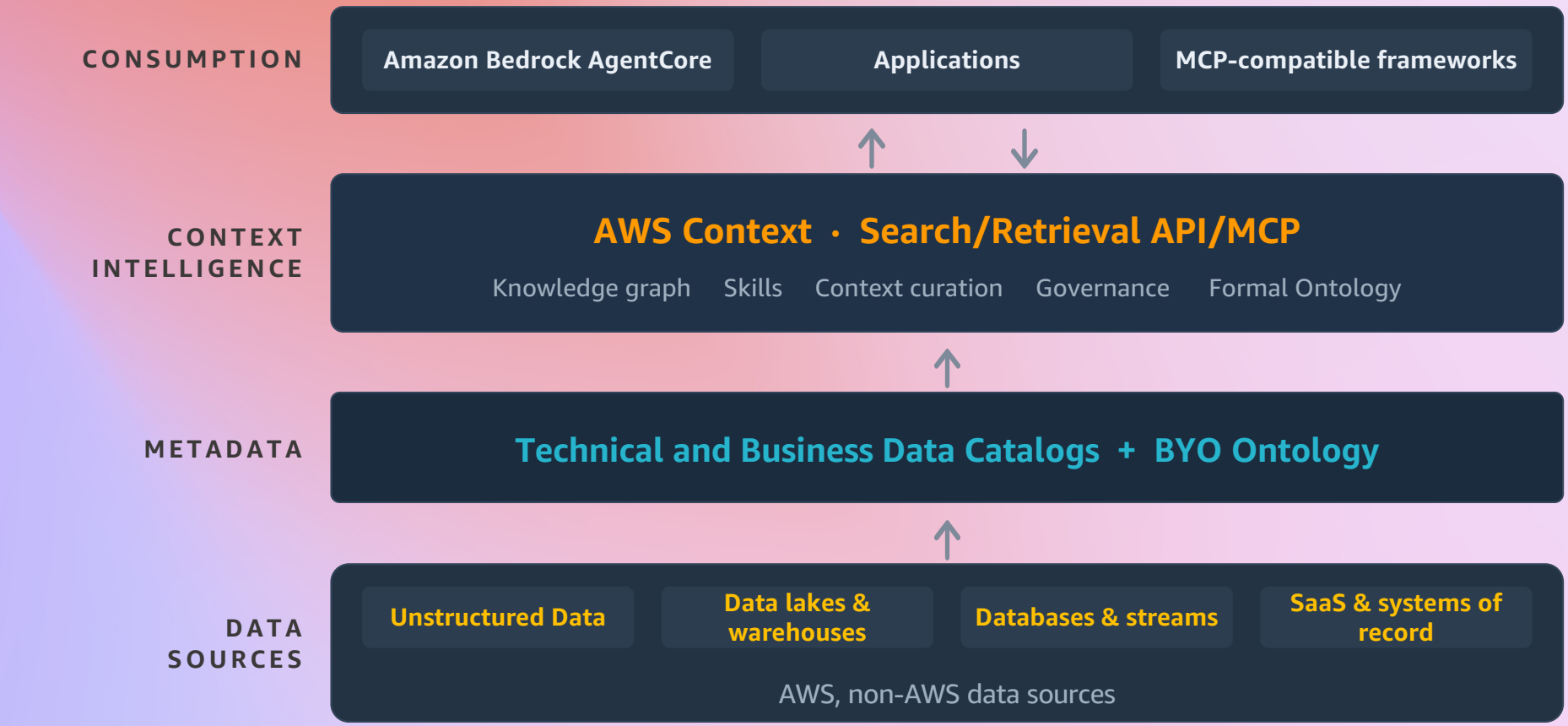
Introducing AWS Context

AWS Context is the full stack that agents reason over: consolidated semantics, knowledge graph, and formal ontologies. Users can start anywhere in the stack, so agents find the right data and reason over it correctly.



aws

Architecture: How AWS Context works



A Flywheel of the Context Intelligence



Example: Customer Trend Insights

Day 1:
Add two datasets, agent finds path: revenue → customer_dim via account_id
Agent only see limited context, can only conclude on local information. Confidence 40% (dashed, uncertain)

Day 3:
Steward linked 40 more datasets, confirms join keys + WBR and marketing reports to guide KPI definition.
Agent understand product vision and access to more organization data estate – confidence 85%

Day 30
47 agents built on this context instance – with Active Daily user of 683.
Company executive provides more guidance through the internal tool; updated emails and market report synced into the context. 76 new skills attached and resulted conflicts – confidence 94%

© 2

Who uses AWS Context

The graph builds itself — these personas govern and consume it.

Data platform teams

Author and promote the graph

- Review inferred relationships in draft
- Promote correct ones to production
- Attach business definitions and Skills
- Monitor how agents use the graph

"I have to be the human API for my own data."

Agent builders

Integrate and consume the graph

- Connect agents via API and MCP
- Agents read governed relationships at runtime
- No per-app context engineering
- One integration serves every agent

"Every agent I build needs to re-learn how our data works."

Data stewards

Define and govern meaning

- Define glossary terms and classifications
- Govern business definitions centrally
- Set usage rules and access policies
- Ensure consistent meaning across teams

"I have no governed, central place to define what our data means."

Core Values

Always-on, self-learning graph

Contexts from the sources are continuously being ingested to the graph. The graph then is evolving based on signals and usages across sources.

No Lock-in, It's Open.

Context will be stored as open table format in your S3. Query it with any compatible engines. You own it.

Compounded Knowledge

Memory and signals from agent trace, together with teammates' clarifications and thumb-ups will be compounded as shared knowledge

Identity-aware governance

Agents only see what user's identity authorizes them to access. Every interaction is auditable through the controls you already use.

aws

RETAIL / CPG

Promo ROI & Demand Sensing: From Weeks to Hours

A global retailer with 2,000+ stores needed to unify promotional effectiveness across channels, replacing fragmented spreadsheets with a single source of truth.

CHALLENGE

- Promotional data siloed across 5 ERPs and 12 regional planning tools
- 6 to 8 week lag between promo execution and ROI visibility
- No shared ontology for product hierarchy, store clusters, or promo types
- Demand sensing models running on stale, inconsistent inputs

SOLUTION WITH CONTEXT & SEMANTIC LAYER

- Unified ontology mapping product, store, and promo hierarchies across all source systems
- Semantic agents reconcile actuals vs. forecast daily, surfacing anomalies automatically
- Allocation recommendations to shift spend mid-campaign
- ROI visibility reduced from weeks to hours

aws

AUTOMOTIVE / MANUFACTURING

Agentic Search over the 20 PB Enterprise Data Lake

A global automotive manufacturer needed AI agents to discover and reason over a petabyte-scale data lake, without years of manual curation or platform lock-in.

CHALLENGE

- Hundreds of thousands of tables and millions of columns spread across business domains
- Cryptic table and column names with little business documentation
- Manual curation at this scale would take years of data steward effort
- Custom agents could see the data but not its meaning, and the team refused vendor lock-in

SOLUTION WITH CONTEXT & SEMANTIC LAYER

- AI-inferred semantics auto-generate business descriptions from schemas and metadata
- Knowledge graph links data assets, domains, and business terms
- Agents on Amazon Bedrock find the right datasets through natural-language search
- Open, customer-owned architecture usable from any agent framework, now in production

aws

Takeaways

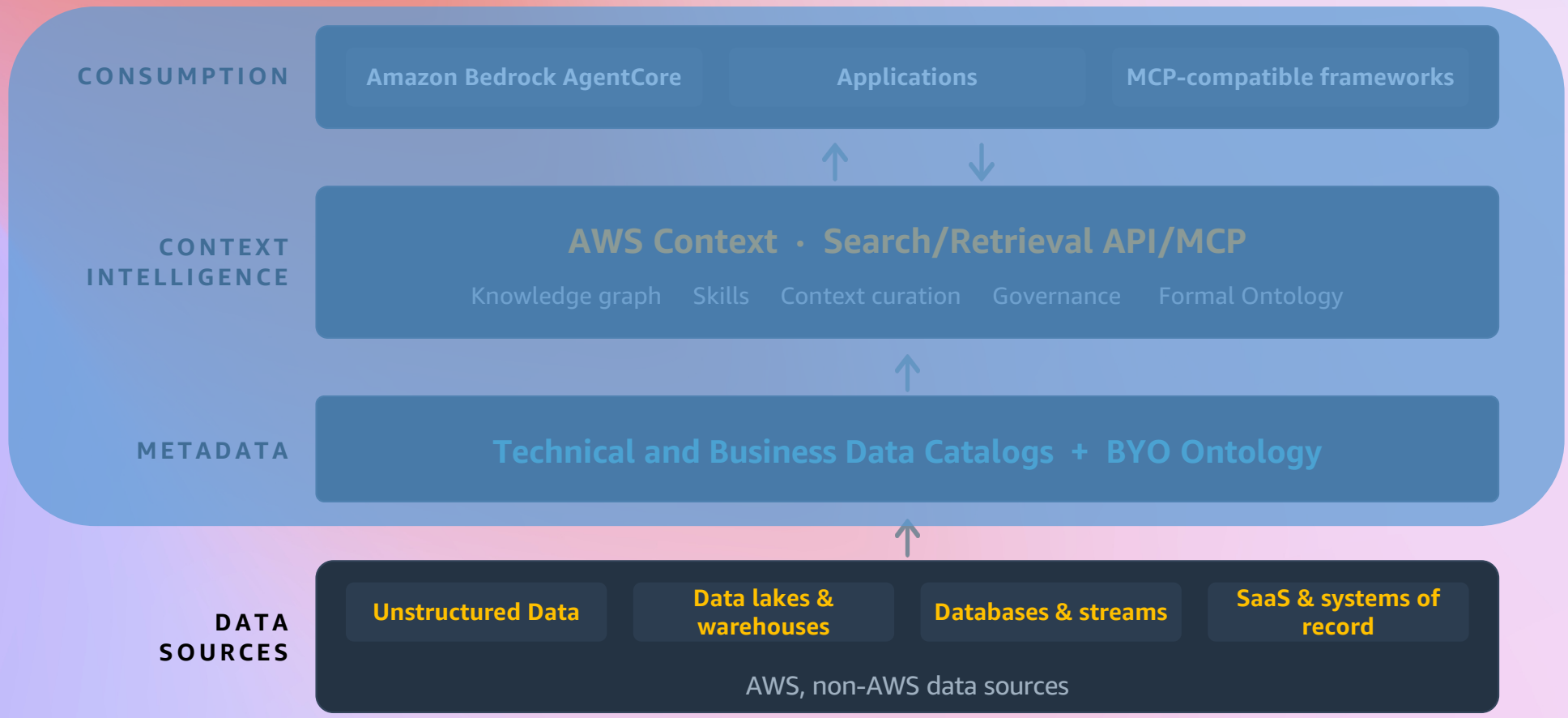
© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.

1. A semantic layer is a **reusable context layer**, not just a data-access path.

2. A semantic layer makes every AI answer **accurate, consistent, and explainable (ACE).**

3. **Semantic layer is the new infrastructure for GenAI.**

Metadata Is Where the Journey Begins



TAKEAWAY

Start Today with Context Ontology Accelerator

Before AWS Context becomes available, you can already give your AI agents business context with the open-source version, built on the same ontology-based approach.

01

Open Source

Apache 2.0 licensed. Deploy it in your own AWS account and own every piece of your business logic.

02

Open W3C Standards

Built on OWL, RDF, and SHACL. Your ontology stays portable, with no proprietary lock-in.

03

Any Agent, via MCP

Serve context to Amazon Bedrock, Strands, LangChain, or any MCP-compatible agent framework.

Get started: github.com/aws/context-ontology-accelerator

aws

Thank You



